

UNIT-II
VECTOR DIFFERENTIATION
TUTORIAL SHEET-1

1. If $\vec{f}$ is a vector function with constant magnitude then $\vec{f} \cdot \frac{d\vec{f}}{dt} = \underline{\hspace{2cm}}$

Ans: 0

2. The displacement of a particle moving along a path is given by $x = (1 - t^3)$, $y = (1 + t^2)$, $z = (2t - 5)$ the magnitude of velocity vector at $t = 1$ second is _____

Ans: $\sqrt{17}$

3. For the curves whose equations are given below, find the unit tangent vectors:

a) $x = t^2 + 1, y = 4t - 3, z = 2(t^2 - 3t)$ at $t = 0$

b) $\vec{r} = (a \cos 3t) \hat{i} + (a \sin 3t) \hat{j} + (4at) \hat{k}$ at $t = \frac{\pi}{4}$

Ans: (i) $\hat{t} = \frac{(2\hat{j} - 3\hat{k})}{\sqrt{13}}$

(ii) $\hat{t} = \frac{1}{5\sqrt{2}} [-3\hat{i} - 3\hat{j} + 4\sqrt{2}\hat{k}]$

4. A particle moves along the curve $\vec{r} = 2t^2 \hat{i} + (t^2 - 4t) \hat{j} + (3t - 5) \hat{k}$. Find the component of velocity and acceleration in the direction of vector $c = i - 3j + 2k$ at $t = 1$.

Ans: $\frac{16}{\sqrt{14}}$ & $\frac{-2}{\sqrt{14}}$

5. A person on a hang glider is spiralling upward due to rapidly rising air on a path having position vector $r(t) = 3 \cos(t) \hat{i} + 3 \sin(t) \hat{j} + t^2 \hat{k}$. Find (a) the velocity and acceleration vectors (b) the glider's speed at any time t .

Ans: $v = 3 \sin(t) \hat{i} + 3 \cos(t) \hat{j} + 2t \hat{k}$;

$a = -3 \cos t \hat{i} - 3 \sin t \hat{j} + 2 \hat{k}$;

$|v| = \sqrt{9 + 4t^2}$



TUTORIAL SHEET-2

1. If $\phi(x, y, z) = xy^2z^3 - x^3y^2z$, then $|\nabla\phi|$ at $(1, -1, 1)$ is _____.
Ans: $2\sqrt{2}$
2. The maximum directional derivative of $\phi(x, y, z) = x^2y + yz^2 - xz^3$ at $(-1, 2, 1)$ is _____.
Ans: $\sqrt{78}$
3. If $\phi(x, y, z) = x^2 + \sin y + z$ then $|\nabla\phi|$ at $(0, \frac{\pi}{2}, 1)$ is _____.
Ans: $\hat{k}$
4. Find the unit normal vector to the surface $\phi(x, y, z) = x^2y + y^2z + xz^2 - 5$ at the point $(1, -1, 2)$.
Ans: $\frac{1}{\sqrt{38}}(2\hat{i} - 3\hat{j} + 5\hat{k})$
5. Find the constants a and b so that the surface $3x^2 - 2y^2 - 3z^2 + 8 = 0$ is orthogonal to the surface $ax^2 + y^2 = bz$ at the point $(-1, 2, 1)$.
Ans: $a = \frac{4}{9}, b = \frac{40}{9}$
6. Find the directional derivative of $\phi(x, y, z) = xyz - xy^2z^3$ at $(1, 2, -1)$ in the direction of $\hat{i} - \hat{j} - 3\hat{k}$.
Ans: $\frac{29}{\sqrt{11}}$

TUTORIAL SHEET-3

1. If $\vec{f} = 3x^2\hat{i} + 5xy^2\hat{j} + xyz^3\hat{k}$ then find $\text{div } \vec{f}$ at $(1, 2, 3)$ is _____.

Ans: 80

2. If $\vec{f} = (y^2 + z^2 - x^2)\hat{i} + (z^2 + x^2 - y^2)\hat{j} + (x^2 + y^2 - z^2)\hat{k}$ then find $\text{div } \vec{f}$ and $\text{curl } \vec{f}$

Ans: $\text{div } \vec{f} = -2(x + y + z)$ $\text{curl } \vec{f} = 2[(y - z)\hat{i} + (z - x)\hat{j} + (x - y)\hat{k}]$

3. Show that the vector field $\vec{f} = (x + 3y)\hat{i} + (y - 3z)\hat{j} + (x - 2z)\hat{k}$ is solenoidal.

4. Determine the constant a such that the vector field $\vec{f} = 3x\hat{i} + (x + y)\hat{j} - az\hat{k}$ is solenoidal.

Ans: 4

5. If $\vec{f} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ and $\vec{g} = yz\hat{i} + xz\hat{j} + xy\hat{k}$ then show that $\vec{f} \times \vec{g}$ is solenoidal.

6. If $\vec{f} = (2x + 3y + az)\hat{i} + (bx + 2y + 3z)\hat{j} + (2x + cy + 3z)\hat{k}$ is irrotational vector field, then find the constants a, b, c .

Ans: $a = 2, b = 3, c = 3$

7. If $\phi = x^2y + 2xy + z^2$, then show that $\nabla\phi$ is irrotational.

8. If $\phi = x^2 - y^2$ then show that ϕ satisfies the Laplacian equation.

9. If $\phi = 2x^2yz^3$ then find $\nabla^2\phi$ at $(1, 1, 1)$.

Ans: 1

10. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$ then show that $r^n\vec{r}$ is irrotational for all values of n and solenoidal for $n = -3$.

11. Show that $\vec{f} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational. Find the function ϕ such that $\vec{f} = \nabla\phi$.

Ans: $\phi = 3x^2y + xz^3 - yz$.